# Preserving lesion quantification in longitudinal whole-body PSMA PET/CT registration

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**Abstract.** **Purpose:** Registering follow-up to baseline PSMA PET/CT scans enables voxel-wise comparison of tumour burden, but the deformation can itself change the metabolic tumour volume (MTV) and total lesion glycolysis (TLG) of lesions. We examine how PET-preservation and bone-rigidity losses trade anatomical alignment for less registration-induced change, and whether the losses are better applied in network training or in instance optimisation (IO).

**Approach:** We trained LapIRN networks with and without PET-preservation and bone-rigidity losses on the Learn2Reg 2026 PSMAReg data, applied the losses in training, IO, or both, and compared against tuned NiftyReg and ConvexAdam on 28 held-out patients, with patient-level statistics and a per-lesion analysis. [TODO: Rigidity penalties compared.]

**Results:** Our IO variants were more accurate than NiftyReg and ConvexAdam (CT Dice 75.6–79.2 % vs. 69.4 % and 61.7 %), with every pair below the folding threshold. With IO and no rigidity loss, adding the PET losses lowered the MTV change from 10.2 % to 6.3 % and the TLG change from 11.9 % to 7.2 % at a cost of 2.8 Dice points. Applying both loss groups only in IO gained 1.7 Dice points over using them in training and IO; the paired MTV and TLG differences were $+0.8$ points (95 % CI $-1.0$ to 2.7) and $+0.2$ points ($-1.9$ to 2.4), respectively. IO removed almost all volume change of its predicted lesion masks; missed lesions contributed most of the remaining measured change. [TODO: Rigidity result.]

**Conclusions:** Applying preservation losses only during IO improved alignment over using them in both training and IO, but equivalent PET preservation was not established. This option requires time for per-pair refinement and predicted lesion masks. Without IO, training with the losses reduced PET changes. The benefit of IO depends on whether its masks cover the lesions being measured.

**Keywords:** deformable image registration, PSMA PET/CT, longitudinal imaging, biomarker preservation, instance optimization.

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## 1 Introduction

Prostate-specific membrane antigen (PSMA) PET/CT is now routinely used for staging prostate cancer, for planning and following radioligand therapy, and for assessing response.[1] Longitudinal imaging provides a baseline examination and one or more follow-up examinations. Registering

each follow-up to the baseline makes it possible to compare tumour burden voxel by voxel, track individual lesions, and accumulate dose.[2] For these uses, a registration must do two things at once: it has to align the anatomy, and it must not itself change the PET-derived quantities that are compared across time points. Two such quantities are the metabolic tumour volume (MTV) and the total lesion glycolysis (TLG). A deformation that compresses or stretches a lesion changes both, so a change measured after registration can be an artefact of the registration rather than of the disease. The Learn2Reg 2026 PSMAReg challenge[3–5] formalises this setting: each follow-up scan is warped to its baseline by a dense displacement field, which is scored on organ overlap, boundary distance, folding, and the change of MTV and TLG caused by the warp. The two goals can compete: limiting deformation may protect lesion measurements while leaving the anatomy misaligned, whereas improving alignment may distort the lesions being measured.

Classical registration solves an optimisation problem per image pair;[6–8] this is accurate but slow for whole-body scans. Learning-based registration replaces the per-pair optimisation with one forward pass of a trained network,[9–11] and instance optimisation (IO) can refine the network output per pair at test time. The coarse-to-fine Laplacian pyramid network LapIRN[12] recovers the large deformations between whole-body scans while keeping the transformation smooth and invertible. With a network and optional IO, preservation constraints can act during training, during per-pair refinement, or at both stages; the choice affects both registration quality and runtime. In the label-guided IO studied here, refinement also requires organ and lesion masks for the scans being registered, which must be predicted when manual labels are unavailable.

Our earlier Learn2Reg submission introduced the affine–LapIRN–IO pipeline with PET-preservation and per-bone rigidity losses and evaluated its selected configurations on challenge validation cases using surrogate labels.[13] In this extension, we retrain on a new patient-level split of the released

labelled cohort, reserve 28 patients for final evaluation, and add controlled loss-placement comparisons, tuned baselines, patient-level statistics, and per-lesion analyses. These 28 patients were among the challenge data available during development of the earlier method, so the split provides a holdout for the present analyses rather than an independent external cohort.

First, we ask whether the PET-preservation and bone-rigidity losses reduce registration-induced change and what alignment accuracy they cost, both with and without IO. We then ask whether applying the losses only during IO can recover the preservation obtained by also using them in training, and compare that choice with the faster network-only setting. Finally, we analyse changes to individual lesions and the coverage of the predicted lesion masks used by IO to understand what limits preservation. Because many PSMA lesions lie in bone, we also compare rigidity penalties for low-resolution whole-body CT, where thin skeletal structures make local constraints difficult to apply. [TODO: Check against the rigidity results.] Together, these experiments test where preservation constraints should act and what limits the benefit of per-pair refinement.

## 2 Methods

### *2.1 Data and Split*

We used the training set of the PSMAReg challenge,[2,5] a longitudinal whole-body PSMA PET/CT cohort. Each scan consists of a CT and a PET image (in SUV), resampled to $2.7344 \times 2.7344 \times 3.27$ mm and cropped to $192 \times 192 \times 288$ voxels. For every scan, the challenge provides 117 CT organ labels (the TotalSegmentator[14] classes) and a PSMA lesion mask. Following the challenge, a follow-up scan is always the moving image and the corresponding baseline scan the fixed image.

We split the 146 labelled patients with at least two scans at the patient level into training, validation, and test sets, aiming at 70/15/15 % of the registration pairs and balancing the sets for

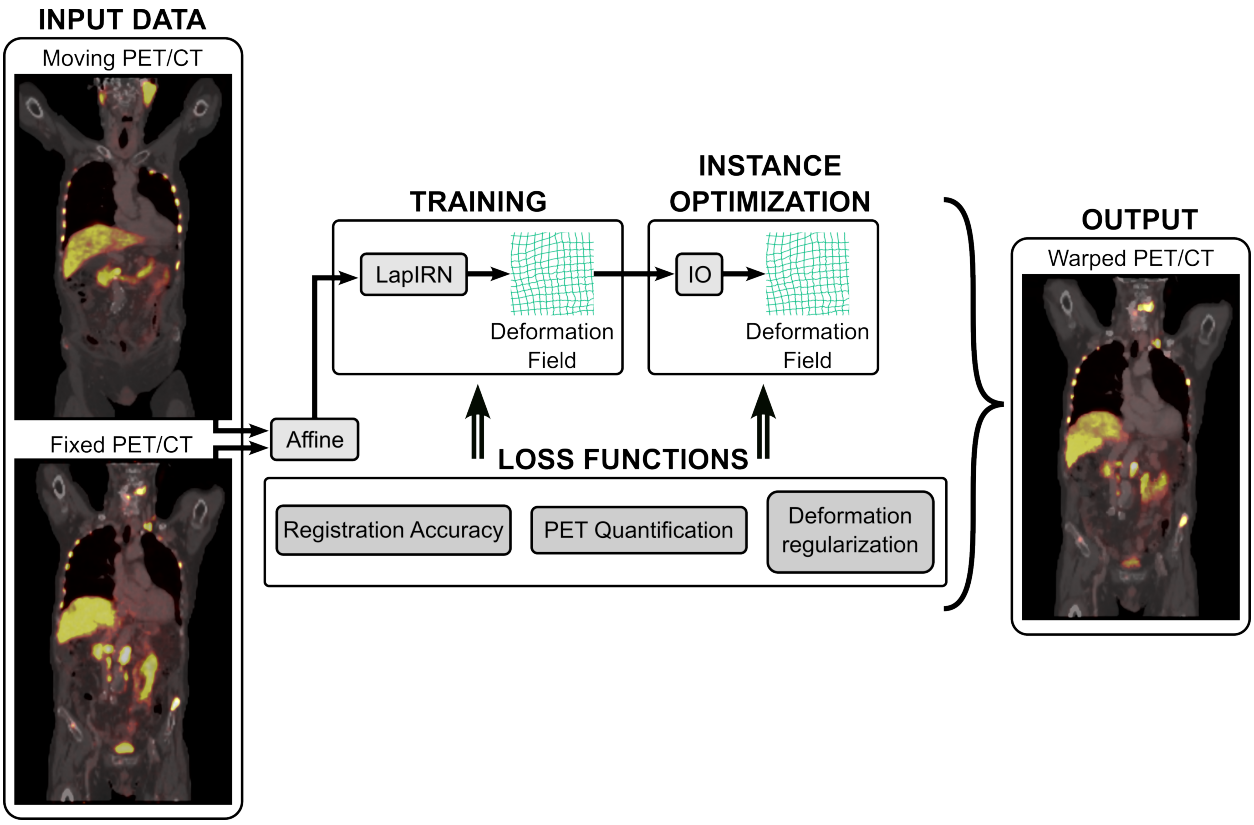



**Fig 1** Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

tumour burden and for bone- versus soft-tissue lesion location. Training uses all pairs of later onto earlier examinations of a patient; validation and testing use only follow-up-to-baseline pairs, as in the challenge. This gave 92/26/28 patients and 178/38/38 pairs. The split was built once with a fixed seed and frozen before any experiment of this study; all choices of settings, checkpoints, and methods were made on the validation set, and the test set was used only for the final evaluation. All patients were part of the challenge training data used to develop the earlier version of the method;[13] the test patients were not used for any decision in the present study.

**Labels used by the method and for evaluation.** Network training uses the provided organ and lesion labels of the training patients. Instance optimisation (Sec. 2.2) also needs organ and lesion labels, but of the scans being registered; to keep the evaluation independent of these inputs, IO uses only predicted labels, and the provided labels of validation and test scans are used only for evaluation (Sec. 3.1). Organ labels are predicted from the CT by TotalSegmentator.[14] Lesion masks are predicted from PET/CT by an nnU-Net[15] set up as the winning autoPET III solution:[16] a residual-encoder U-Net (ResEnc-L) with $192^3$ patches, initialised from their pretrained weights, with an auxiliary organ-segmentation head and misalignment augmentation. It was trained with patient-grouped five-fold cross-validation on all 494 scans of the non-test patients, including patients with a single examination. Validation scans are segmented by the fold model that did not see the patient, and test scans by the ensemble of the five fold models. The predicted lesion masks reach a mean Dice of 0.56 on the validation scans and 0.51 on the test scans (medians 0.69 and 0.67).

**Preprocessing.** For the network and IO, CT was clipped to $[-1000, 1500]$ HU and PET to $[0, 20]$ SUV, and both were rescaled to $[0, 1]$. As in the ConvexAdam baseline of the challenge,[17] the patient bed was removed from the CT. All PET quantities in the evaluation are computed on the original SUV images.

**PET measures.** The PSMAReg challenge uses metabolic tumour volume (MTV) and total lesion glycolysis (TLG) to measure how registration changes the moving scan's lesion volume and uptake.[2] For a PET image $P$ in SUV, binary lesion mask $M$, and voxel volume $v$, they are

$$\mathrm{MTV}(M) = v \sum_x M(x), \qquad \mathrm{TLG}(P, M) = v \sum_x P(x) M(x). \tag{1}$$

Both quantities are computed on the same moving scan before and after warping it to the fixed frame; the challenge scorer warps PET linearly and the lesion mask by nearest neighbour. The reported preservation error for $Q \in \{\mathrm{MTV}, \mathrm{TLG}\}$ is its absolute relative change,

$$|\Delta Q| = \frac{|Q_{\text{after}} - Q_{\text{before}}|}{\max(Q_{\text{before}}, 10^{-6})}. \tag{2}$$

This measures change induced by registration, not change in disease burden between examinations.

### *2.2 Registration Pipeline*

The pipeline (Fig. 1) has three stages, all estimated from the moving scan $(C_m, P_m)$ and the fixed scan $(C_f, P_f)$, where $C$ denotes CT and $P$ PET. We denote the affine transform by $\varphi_{\text{aff}}$, the network deformation by $\varphi_\theta$, and their composition, which maps the original moving scan to the fixed scan, by $\varphi$.

**Affine pre-registration.** ANTs[18] estimates an affine transform from the two CTs, using body masks, a $[-1000, 1000]$ HU window, half-resolution images, and Mattes mutual information. It is converted to a dense displacement field.

**Deformable network.** We use LapIRN[12] with three pyramid levels at quarter, half, and full resolution. Each level predicts a stationary velocity field that is added to the upsampled velocity of the coarser level and integrated by scaling and squaring[19] into $\varphi_\theta$. Each level receives four channels: the fixed CT and PET and the affinely aligned moving CT and PET. The levels are trained one after the other; a finer level is initialised from the coarser one, which is first kept frozen and then trained jointly. Each sub-network has 24 base channels and a trunk of ten residual blocks with an expansion factor of two.

**Instance optimisation.** At test time, the deformation of each pair is refined by optimising a residual stationary velocity field, initialised at zero and integrated by scaling and squaring (seven steps), on top of the total transformation $\varphi$. The residual is optimised at full resolution with Adam[20] (learning rate $0.0125$, 40 iterations). The objective is the same as in training (Sec. 2.3), with the predicted labels in place of ground-truth labels. Of the 40 iterates, we return the one with the lowest objective, in which the interpolated (soft) MTV and TLG terms are replaced by their nearest-neighbour (hard) values, i.e. by how the evaluation counts them; this selection uses only predicted labels.

### 2.3 Registration Objective

Network training and IO minimise a weighted sum of three groups of terms:

$$\mathcal{L} = \underbrace{\lambda_{\mathrm{NCC}}\mathcal{L}_{\mathrm{NCC}} + \lambda_{\mathrm{DSC}}\mathcal{L}_{\mathrm{DSC}}}_{\text{alignment}} + \underbrace{\lambda_{\mathrm{MTV}}\mathcal{L}_{\mathrm{MTV}} + \lambda_{\overline{J}}\mathcal{L}_{\overline{J}} + \lambda_{J}\mathcal{L}_{J} + \lambda_{\mathrm{TLG}}\mathcal{L}_{\mathrm{TLG}}}_{\text{PET preservation}} + \underbrace{\lambda_{\mathrm{rig}}\mathcal{L}_{\mathrm{rig}} + \lambda_{\mathrm{smooth}}\mathcal{L}_{\mathrm{smooth}} + \lambda_{\mathrm{NDV}}\mathcal{L}_{\mathrm{NDV}}}_{\text{regularisation}} . \tag{3}$$

All terms are evaluated on the total transformation $\varphi$. The PET-preservation terms and the rigidity term are the subject of this study; they can be switched on in network training, in IO, or in both (Sec. 3). The two coarse network levels never use them, because lesions and thin bones span only a few voxels at these resolutions.

**Alignment.** $\mathcal{L}_{\mathrm{NCC}}$ is the negative local normalised cross-correlation of the CTs, summed over one to three scales depending on the pyramid level. PET is not used for similarity, because uptake changes between examinations with disease and therapy and a PET similarity would reward matching these changes. $\mathcal{L}_{\mathrm{DSC}}$ is a soft Dice loss on the CT organ labels.

**PET-preservation.** For the differentiable loss terms, let $M$ be the moving lesion mask and $\widetilde{M}$, $\widetilde{P}$ the linearly warped mask and PET. Nearest-neighbour warping, as used in the evaluation, is piecewise constant in the displacement and gives no useful gradient; linear warping gives a soft lesion boundary instead. The volume and uptake terms penalise the relative change defined above,

using the soft mask,

$$\mathcal{L}_{\mathrm{MTV}} = \left( \frac{\left| \sum_x \widetilde{M}(x) - \sum_x M(x) \right|}{\sum_x M(x) + \epsilon} \right)^2, \qquad \mathcal{L}_{\mathrm{TLG}} = \frac{\left| \sum_x \widetilde{P}(x)\widetilde{M}(x) - \sum_x P_m(x)M(x) \right|}{\sum_x P_m(x)M(x) + \epsilon}, \tag{4}$$

with $\epsilon = 10^{-5}$. Two further terms act on the Jacobian determinant $\det J_\varphi$ inside the warped lesion region $W$ in the fixed frame:

$$\mathcal{L}_{\overline{J}} = \left( \frac{1}{|W|} \sum_{x \in W} \det J_\varphi(x) - 1 \right)^2, \qquad \mathcal{L}_J = \frac{1}{|W|} \sum_{x \in W} \big( \det J_\varphi(x) - 1 \big)^2. \tag{5}$$

$\mathcal{L}_{\overline{J}}$ constrains the net volume change of all lesions, and $\mathcal{L}_J$ additionally prevents compression in one part of a lesion from being compensated by expansion in another. For $\mathcal{L}_{\overline{J}}$, $W$ is the linearly warped mask (without gradient through the mask), for $\mathcal{L}_J$ the nearest-neighbour warped mask.

**Bone rigidity.** Many PSMA lesions lie in bone, which moves rigidly between examinations. For each of the 61 skeletal labels of the fixed scan (in IO: the predicted labels), we fit a rigid transformation to the displacement of the structure in closed form[21,22] and penalise the mean squared residual of its voxels; $\mathcal{L}_{\mathrm{rig}}$ is the mean over structures with at least 50 voxels. Each bone is fitted independently, so neighbouring bones may move relative to each other, and the term involves no finite differences, so it does not constrain the tissue around a bone.

**Regularisation.** $\mathcal{L}_{\mathrm{smooth}}$ is the mean squared spatial gradient of the displacement. $\mathcal{L}_{\mathrm{NDV}}$ is a differentiable version of the challenge's non-diffeomorphic volume, which splits every voxel into tetrahedra and accumulates the negative part of their Jacobian determinants.[23]

**Table 1** Loss weights of Eq. (3) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

| | $\lambda_{\mathrm{NCC}}$ | $\lambda_{\mathrm{DSC}}$ | $\lambda_{\mathrm{MTV}}$ | $\lambda_{\overline{J}}$ | $\lambda_{J}$ | $\lambda_{\mathrm{TLG}}$ | $\lambda_{\mathrm{rig}}$ | $\lambda_{\mathrm{smooth}}$ | $\lambda_{\mathrm{NDV}}$ |
|---|---|---|---|---|---|---|---|---|---|
| Training | 5 | 5 | 5 | 0.125 | 1.25 | 1.25 | 0.05 | 10 | $10^4$ |
| IO | 5 | 5 | $4 \cdot 10^4$ | $10^3$ | 4 | 16 | 0.25 | 0 | 10 |

**Weights.** Table 1 lists the weights. In training, the alignment weights are the same for all models, and the PET and rigidity weights are either used or set to zero. The IO weights were tuned on the validation set (Sec. 3.3).

### *2.4 Rigidity Penalties Compared*

Besides the per-bone rigid fit, we evaluate the rigidity penalty of Staring et al.,[24] which requires the Jacobian $J$ inside a bone to be a rotation: $\det J = 1$, $J^\top J = I$, and vanishing second derivatives of the deformation. It is computed with finite differences, whose stencil reads the 6 face and 12 edge neighbours of a voxel. At a bone boundary, part of the stencil lies outside the bone, so the penalty also acts on the surrounding soft tissue or on a neighbouring bone. A common remedy is to erode the bone mask so that the stencil stays inside; we erode each bone by exactly this 18-neighbour footprint, which removes the leakage completely. The three variants (Staring, Staring with erosion, per-bone rigid fit) are compared in IO (Sec. 3).

### *2.5 Training*

Each level was trained with Adam[20] for $10^5$, $10^5$, and $1.2 \cdot 10^5$ steps (quarter, half, full resolution) with learning rates of $1.5 \cdot 10^{-4}$, $1.2 \cdot 10^{-3}$, and $10^{-5}$, a batch size of one with gradients accumulated over four pairs, a linear warm-up over five epochs, and a cosine decay to 10 % of the initial rate. A finer level is trained jointly with the coarser ones after ten epochs. Convolutions run in bfloat16,

all losses in single precision. Training took 18 h, 22 h, and 3.7 days for the three levels on one NVIDIA A100 (80 GB). The augmentations mimic longitudinal variability: left–right flipping ($p = 0.5$), CT intensity shifts of $\pm 0.02$ in the normalised window (about $\pm 50$ HU), CT scaling between 0.9 and 1.1, PET scaling between 0.85 and 1.15, and field-of-view mismatch by cropping up to 40 axial slices from both scans and up to 10 further slices from each scan separately. The two coarse levels are shared by all models; the variants of this study differ only in the losses of the full-resolution level. Checkpoints were selected on the validation set: the coarse levels by CT Dice, the full-resolution level by the best accuracy (CT Dice and HD95) among checkpoints whose validation folding stays below $1.4 \cdot 10^{-4}\,\%$ non-diffeomorphic volume. We did not select on MTV or TLG, so that the PET losses are not credited with effects of the checkpoint choice.

**Use of AI tools.** Parts of the code for training, evaluation, statistical analysis, and figure generation were written with the help of an AI coding assistant. All generated code was reviewed and tested by the authors, who are responsible for the analyses and results.

## 3 Experiments

### *3.1 Evaluation Metrics*

All metrics are computed with the provided labels of the test scans, after warping the moving scan with the scorer of the challenge.

**Alignment.** CT Dice and the 95th percentile of the Hausdorff distance (HD95), averaged over the 117 organ labels.

**PET-preservation metrics.** We compute $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ as defined in Eq. (2), using the provided moving-scan lesion masks and the challenge scorer. Because PET and mask are warped separately, the scorer changes TLG even for the identity transformation: on the test pairs with lesions, an all-zero displacement changes TLG by $2.9 \pm 1.2\,\%$ (MTV: $0\,\%$). This resampling floor applies to every warped method; we therefore report the unregistered scans ("Initial") without resampling.

**Per-lesion change.** The aggregate errors of Eq. (2) can hide large changes of individual lesions that cancel. We therefore also split the moving lesion mask into 26-connected components and compute $|\Delta\mathrm{MTV}|$ for each. Lesions are grouped by size (small $< 0.5$ ml, medium $< 5$ ml, large) and location (bone-associated if at least half of the lesion lies on a skeletal label); lesions with fewer than five voxels are excluded.

**Deformation quality.** Percentage of non-diffeomorphic volume (NDV) inside the body, as defined by the challenge.[23]

**Bone metrics.** For the rigidity comparison, we additionally compute Dice and HD95 separately over bones and soft-tissue organs; the distortion of within-bone distances, i.e. the mean relative change of the physical distance between randomly sampled pairs of interior points (eroded mask) of the same bone, separately for short (5–15 mm) and long ($\geq 15$ mm) distances (short distances reflect local compression or bending, long ones the deformation of a bone as a whole; below 5 mm, i.e. about two voxels, the relative change is dominated by interpolation and discretisation); NDV inside bones and in a three-voxel band around them; and, as a secondary measure, the residual of a rigid fit to each bone's displacement (bone rigidity error). We also measure the fraction of stencil

reads that leave the bone of the centre voxel (stencil leakage) and the fraction of bone voxels and of bones that remain after erosion.

### *3.2 Statistical Analysis*

A patient may contribute several pairs and many lesions. We therefore average every metric over the pairs of a patient first and use patients as the unit of all tests, rankings, and confidence intervals (28 test patients). Paired differences between two methods are tested with the two-sided Wilcoxon signed-rank test[25] on patient values and corrected for multiple comparisons with Holm's method[26] within each table and metric; we call a difference significant if the corrected $p < 0.05$. Confidence intervals are 95 % percentile intervals over patient bootstrap samples;[27] for per-lesion statistics, patients are resampled together with all their lesions (cluster bootstrap).

For the comparison with baselines, we additionally rank all methods with the significance-based ranking of the challenge: per metric, every pair of methods is compared with a one-sided Wilcoxon test (Holm-corrected), and a method scores (number of methods it significantly outperforms $+1$)/$K$ for $K$ methods. Scores are averaged into accuracy (CT Dice, HD95), PET preservation ($|\Delta\mathrm{MTV}|$, $|\Delta\mathrm{TLG}|$), and regularity (NDV, after subtracting a practical-equivalence threshold of 0.005 % per pair), and combined into $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$. The challenge additionally scores Dice and HD95 of organs with physiological PET uptake, for which no labels are available to us; we omit them. Dice or HD95 of lesions would not be meaningful either, because lesions appear, disappear, and change between examinations. The stability of the ranks is shown by 95 % rank intervals over 1000 patient bootstrap samples.

### *3.3 Tuning and Model Selection*

All tuning used the validation set and the challenge ranking above as the selection rule: the candidate settings of one grid are ranked against each other, and the top-ranked one is kept.

- **IO:** on the network trained with both losses, step-wise: first learning rate $\times$ iterations ($\{0.0125, 0.025, 0.05, 0.1\} \times \{10, 20, 40\}$), then a common scale of the four PET weights ($\times\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$ of the training-derived values). The ranking does not reward rigidity (its cost in Dice is significant, while folding stays below the threshold for every weight), so it would set $\lambda_{\text{rig}}$ to zero; we fixed $\lambda_{\text{rig}} = 0.25$ instead, which costs 0.3 Dice points on validation.

- **NiftyReg**:[28,29] `reg_aladin` followed by `reg_f3d` on the GPU with normalised mutual information (local NCC is not available on the GPU), over the bending-energy weights $\{0.0002, 0.0005, 0.001, 0.005, 0.02\}$ and final control-point spacings of $\{3, 5, 7, 9\}$ voxels; chosen: $0.0002$ and 9 voxels.

- **ConvexAdam**:[17] after the same ANTs affine as our method, over the regularisation weight $\lambda \in \{0.1, 0.25, 0.5, 1.25, 2, 4\}$; chosen: 0.1.

Several choices ended at the edge of their grid; extending the grids changed the validation scores only marginally.

**Choice of the full method.** Before the experiments, we defined our full method as the network trained with both losses followed by IO with both losses. To check this choice, we also ranked all variants of Sec. 3.4 on the validation set. This ranking favoured the variant without any of the losses, because accuracy differences between variants are significant while differences in MTV

and TLG rarely are. We therefore report both in the comparison with baselines, together with the variant that uses the losses only in IO.

### *3.4 Experiments*

**Effect of the losses.** Four networks were trained with the same alignment and regularisation terms and with neither, the PET-preservation, the rigidity, or both groups of losses. Each is evaluated as network output and with IO using the same losses as in its training.

**Losses in training or only in IO.** The network trained without the losses is combined with IO using PET preservation, rigidity, or both, and compared with the networks trained with these losses (and IO with the same losses). To measure what the network contributes, IO with both losses is also run directly after the affine pre-registration.

**Contribution of each PET term.** On the network trained without the losses, IO with all four PET terms is compared with IO in which one term at a time is switched off.

**Comparison with baselines.** Initial alignment, the affine pre-registration, NiftyReg, Convex-Adam, IO after the affine, and our variants are compared on the test set. Runtimes are per pair on one NVIDIA RTX A6000 GPU and include the affine step.

**Rigidity penalties.** On the network trained with the PET losses, IO with the PET terms is combined with no rigidity, Staring, Staring with erosion, or the per-bone rigid fit. Because the penalties have different units, a reference weight is first calibrated for each so that its initial value on five validation pairs equals that of the rigid fit at $\lambda_{\mathrm{rig}} = 0.25$ (Staring 0.66, Staring with erosion 1.03). Each method is then evaluated on the validation set at $\{0.1, 0.3, 1, 3, 10\}$ times its reference weight,

giving a curve of CT Dice against within-bone distance distortion. For the test set, we choose for each method the largest weight whose validation Dice is within one Dice point of the variant without rigidity. Where the largest weight still met this criterion (Staring with erosion and the rigid fit), we extended the grid by $\times 30$ and $\times 100$ and capped it there.

**Accuracy–preservation trade-off.** On the network trained without the losses, IO with the PET terms (without rigidity) is run with all four PET weights scaled by $\{0, 10^{-4}, 3{\cdot}10^{-4}, 0.001, 0.003, 0.01, 0.03, 0.1, 0.$ relative to the training-derived values; $\times 0.1$ is the setting chosen by the IO tuning. Besides the evaluation on the ground-truth labels, we compute CT Dice, $|\Delta\mathrm{MTV}|$, and $|\Delta\mathrm{TLG}|$ on the predicted labels that IO optimises, with the same definitions. To separate the effect of the predicted lesion masks from that of the optimisation, IO is also run with the ground-truth lesion masks instead of the predicted ones (organ labels remain predicted), at the tuned and at the full PET weights.

**Missed and detected lesions.** Each ground-truth lesion of the moving scan is assigned to a group by the fraction of its voxels covered by the predicted lesion mask: missed (0 %), partly covered ($< 50\,\%$), or covered ($\geq 50\,\%$). For each group, we report two quantities, both computed per scan as in Eq. (2) and averaged over patients: the change within the group, using only the group's lesions, and the contribution of the group to the evaluation metric, i.e. the change if only the group's lesions changed and all other lesions kept their volume and uptake. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more than the evaluation metric.

## 4 Results

All results are on the test set (28 patients, 38 pairs); values are mean±SD over patients, and differences are given in percentage points. We first test whether the losses reduce changes to lesion

measurements and where they should act in the registration pipeline. We then examine individual lesions and PET loss terms, compare with baseline methods, assess alternative rigidity penalties and the accuracy–preservation trade-off, and relate the remaining error to predicted lesion-mask coverage.

### *4.1 Effect of the PET-Preservation and Rigidity Losses*

Table 2 compares four configurations: no PET-preservation or bone-rigidity loss, PET preservation alone, rigidity alone, and both. Each network is evaluated before and after IO. In IO, the PET-preservation and bone-rigidity losses are enabled or disabled to match that network's training configuration. Training with the PET losses lowered the volume change of the lesions from 10.6 % to 7.6 % and the uptake change from 10.8 % to 7.6 %, at the cost of 1.9 Dice points. IO raised the Dice of every network by 2.8 to 4.4 points and kept the effect of the losses: with the PET losses, $|\Delta\mathrm{MTV}|$ was 6.3 % instead of 10.2 % and $|\Delta\mathrm{TLG}|$ 7.2 % instead of 11.9 %, for 2.8 Dice points. The rigidity loss alone did not change the Dice of the network and cost 0.5 points with IO, and it lowered $|\Delta\mathrm{MTV}|$ in both cases (network 9.4 %, with IO 7.9 %). Combining both groups changed the mean PET metrics little relative to the PET losses alone and was the least accurate variant (72.8 % and 75.6 % Dice). HD95 followed the Dice in every case (network 7.2 to 7.5 mm, with IO 6.8 to 7.2 mm). No variant folded: the non-diffeomorphic volume of every pair stayed below the 0.005 % threshold of the challenge (at most 0.6 ppm on average).

The losses therefore change the accuracy–preservation balance; the next question is whether they need to shape the trained network or can act only during per-pair refinement.

**Table 2** Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean±SD over patients. †/‡: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

| | Network | | | Network + IO | | |
|---|---|---|---|---|---|---|
| Losses | Dice (%) ↑ | $\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓ | $\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓ | Dice (%) ↑ | $\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓ | $\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓ |
| Neither | 74.8±4.5 | 10.6±10.2 | 10.8±11.7 | 79.2±4.3 | 10.2±10.1 | 11.9±12.5 |
| PET | 72.9±4.4‡ | 7.6±9.0† | 7.6±9.3† | 76.4±4.7‡ | 6.3±9.1† | 7.2±9.4† |
| Rigidity | 74.8±4.5 | 9.4±9.5† | 9.5±10.9† | 78.7±4.2‡ | 7.9±9.0† | 10.1±11.0 |
| PET + rigidity | 72.8±4.3‡ | 7.4±9.0† | 7.7±10.1† | 75.6±4.8‡ | 6.2±8.7† | 7.9±9.0 |

### *4.2 Losses in Training or Only in IO*

To test whether the losses need to act during training, Fig. 2 compares networks trained with each loss group against the network trained without them, followed by IO with the same losses in each comparison. With both loss groups, applying them only in IO increased Dice by 1.7 points (95 % CI 1.1 to 2.2) compared with using them in training and IO. The paired differences (IO only minus training and IO) were $+0.8$ points in $|\Delta\mathrm{MTV}|$ (95 % CI $-1.0$ to 2.7) and $+0.2$ points in $|\Delta\mathrm{TLG}|$ ($-1.9$ to 2.4). With PET losses alone, the Dice gain was 1.2 points (0.4 to 1.9), while $|\Delta\mathrm{MTV}|$ and $|\Delta\mathrm{TLG}|$ increased by 1.3 (0.0 to 2.6) and 1.5 ($-0.1$ to 3.0) points. With rigidity alone, the paired differences were 0.0 ($-0.1$ to 0.1) Dice points, $+0.5$ (0.0 to 1.0) MTV points, and $+0.2$ ($-0.4$ to 0.8) TLG points. None of the preservation differences was significant after Holm correction; the intervals do not establish equivalent preservation. Without IO, training with the losses reduced registration-induced PET changes in the network-only variants tested (Table 2, left).

### *4.3 Per-Lesion Volume Change*

To reveal changes hidden by scan-level aggregation, Fig. 3 shows the volume change of 1327 individual lesions (26 patients with lesions). Small lesions changed most in every condition: for bone-associated lesions with IO and without the losses, the median change was 33 % (small), 17 % (medium), and 9 % (large). A small lesion has fewer than about 20 voxels, so gaining or losing

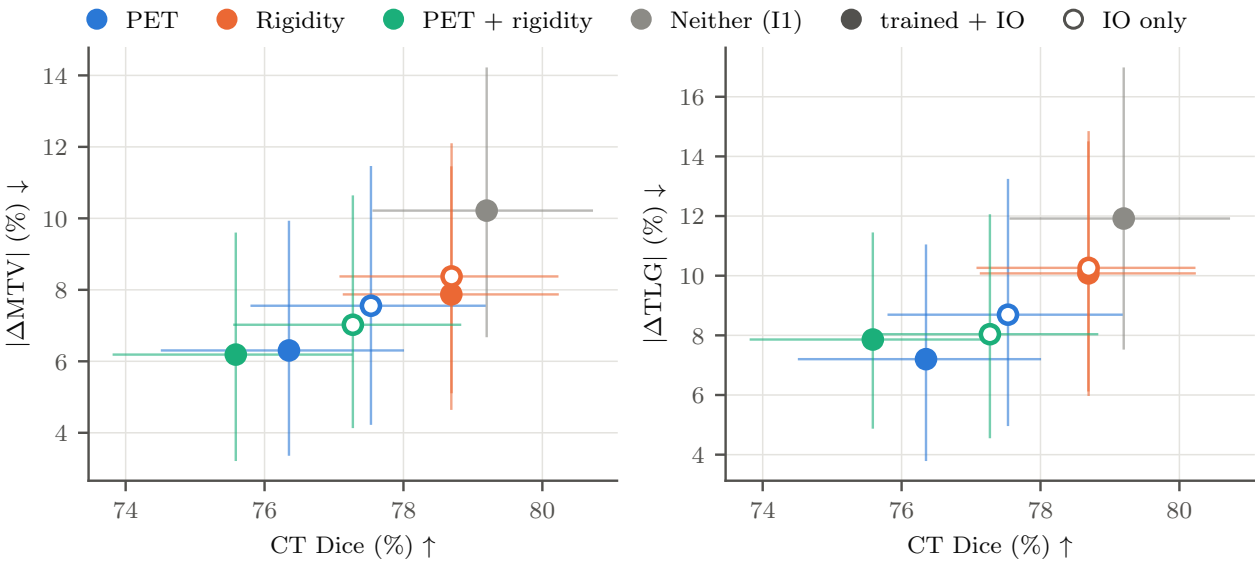



**Fig 2** Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small bone lesions (median from 20 % to 33 %); bone lesions tended to grow (median signed change $+3$ to $+10$ %) and soft-tissue lesions to shrink ($-9$ to $-13$ %). The rigidity loss acted only on bone-associated lesions, as expected from its construction: with IO, it lowered the median change of small, medium, and large bone lesions from 33 % to 20 %, 17 % to 12 %, and 9 % to 5 %, and left soft-tissue lesions unchanged (17 to 22 %). The PET losses acted in both locations, for example from 18 % to 10 % for medium soft-tissue lesions and from 33 % to 13 % for small bone lesions; only for small soft-tissue lesions, the smallest group (41 lesions), no clear effect was visible. With both groups, bone lesions changed least at every size (10 %, 9 %, and 4 %), and their systematic growth disappeared (median signed change about 0). The medians of small lesions take only a few distinct values (e.g. 10.0 %, 12.5 %, 20.0 %) because they are ratios of small voxel counts.

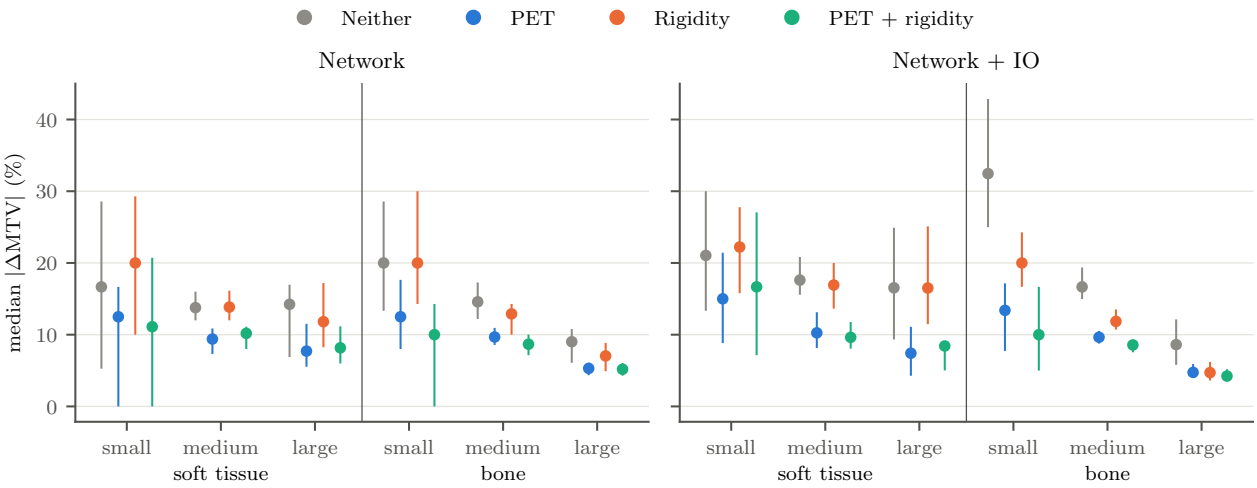



**Fig 3** Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, $\geq$50 % of the lesion on bone) and size (small $<$0.5 ml, medium $<$5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

**Table 3** Contribution of each PET term in IO (test set, 28 patients): network trained without PET losses (T1), IO with all PET terms or with one term removed. Mean$\pm$SD over patients. $^{\dagger}$/$^{\ddagger}$: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

| IO PET terms | Dice (%) ↑ | $\lvert\Delta\mathrm{MTV}\rvert$ (%) ↓ | $\lvert\Delta\mathrm{TLG}\rvert$ (%) ↓ |
|---|---|---|---|
| All PET terms | 77.5±4.5 | 7.6±10.0 | 8.7±11.1 |
| without $\mathcal{L}_{\mathrm{MTV}}$ | 78.8±4.3$^{\dagger}$ | 8.0±9.8 | 8.2±11.0 |
| without $\mathcal{L}_{\overline{J}}$ | 77.6±4.6 | 8.3±9.9$^{\ddagger}$ | 8.9±11.2 |
| without $\mathcal{L}_{\mathrm{TLG}}$ | 77.5±4.5 | 7.2±10.1 | 9.1±10.8 |
| without $\mathcal{L}_{J}$ | 77.6±4.5$^{\dagger}$ | 7.8±9.9$^{\ddagger}$ | 9.1±10.9 |

### *4.4 Contribution of Each PET Term*

To identify which constraints account for preservation in IO, we remove one PET term at a time from IO on the network trained without the losses (Table 3). The two Jacobian terms carried the volume preservation: without $\mathcal{L}_{\overline{J}}$, $|\Delta\mathrm{MTV}|$ increased from 7.6 % to 8.3 %, and without $\mathcal{L}_{J}$ to 7.8 % (both significant). The volume term $\mathcal{L}_{\mathrm{MTV}}$ mainly cost accuracy: without it, Dice increased by 1.3 points (from 77.5 % to 78.8 %, significant), while $|\Delta\mathrm{MTV}|$ increased only from 7.6 % to 8.0 % and $|\Delta\mathrm{TLG}|$ decreased from 8.7 % to 8.2 % (neither significant). Switching off $\mathcal{L}_{\mathrm{TLG}}$ changed no metric significantly ($|\Delta\mathrm{TLG}|$ 9.1 %, $|\Delta\mathrm{MTV}|$ 7.2 %). No single term changed $|\Delta\mathrm{TLG}|$ significantly, and all changes of $|\Delta\mathrm{MTV}|$ were below one percentage point.

### *4.5 Comparison with Baselines*

To place these operating points in context, we compare our variants with established registration methods in Table 4 and Fig. 4. Every variant of ours was significantly more accurate than NiftyReg (69.4 % Dice, 8.6 mm HD95), ConvexAdam (61.7 %, 9.2 mm), and the affine pre-registration (46.4 %, 13.9 mm), and none of them folded. NiftyReg, tuned on the validation set where it rarely folded, exceeded the folding threshold on 21 % of the test pairs (mean 1170 ppm, driven by three pairs with 0.8 to 1.3 %). IO directly after the affine pre-registration reached only 52.7 % Dice: within the same iteration budget, IO cannot replace the network. Its MTV and TLG change was the lowest of all methods (3.4 %), because the deformation it adds to the affine is small.

Among our variants, IO without the PET and rigidity losses was the most accurate (79.2 % Dice), and our full method, with the losses in training and IO, preserved best ($|\Delta\mathrm{MTV}|$ 6.2 %, significantly lower than without the losses). The variant with the losses only in IO ranked first (77.3 % Dice, 7.0 % $|\Delta\mathrm{MTV}|$), followed by the full method and the variant without the losses; their rank intervals overlap (1–5). The ranking was decided mainly by accuracy: after correction for multiple comparisons, few differences in preservation were significant (on $|\Delta\mathrm{MTV}|$, each variant with the losses outperformed one other method and IO after the affine two; on $|\Delta\mathrm{TLG}|$, only IO after the affine outperformed others), and none in folding. The network alone needs 11 s per pair, as fast as ConvexAdam and ten times faster than NiftyReg (108 s); IO adds about 2.5 minutes. Figure 5 shows a test pair whose lesion volume grew tenfold between baseline and follow-up (from 2400 to 24 191 voxels). The scans were initially far apart (CT Dice 3.5 %); our full method aligned them to 81.0 % Dice (NiftyReg 69.5 %, ConvexAdam 65.6 %), while the volume and uptake of the 163 follow-up lesions changed by only $-0.1\,\%$ and $+0.9\,\%$ and no voxel folded. Before registration,

**Table 4** Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). > thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

| Method | Training | IO | Rank ↓ | Dice (%) ↑ | HD95 (mm) ↓ | \|ΔMTV\| (%) ↓ | \|ΔTLG\| (%) ↓ | NDV (ppm) ↓ | > thr. (%) | Runtime (s) |
|---|---|---|---|---|---|---|---|---|---|---|
| Initial | – | – | – | 9.5±8.1† | 44.37±21.32† | 0.0±0.0 | 0.0±0.0 | 0±0 | 0 | – |
| Affine | – | – | 9 [8–9] | 46.4±10.8† | 13.92±4.74† | 4.6±4.6 | 5.7±3.9 | **0**±0 | 0 | 10.4±0.3 |
| NiftyReg | – | – | 7 [4–7] | 69.4±4.7† | 8.60±3.69† | 7.6±12.5 | 11.3±13.7 | 1170±3120 | 21 | 107.7±24.9 |
| ConvexAdam | – | – | 8 [7–8] | 61.7±4.5† | 9.16±3.14† | 9.3±12.0 | 9.6±15.8 | **0**±0 | 0 | 11.3±0.6 |
| Affine + IO | – | PET + rig. | 5 [1–8] | 52.7±12.1† | 13.26±4.77† | **3.4**±3.2 | **3.4**±3.6 | 0.1±0.099 | 0 | 155.5±2.6 |
| Ours | – | – | 6 [3–7] | 74.8±4.5† | 7.23±2.32 | 10.6±10.2† | 10.8±11.7 | 0.0037±0.01 | 0 | 11.0±0.4 |
| Ours | PET + rig. | – | 4 [1–6] | 72.8±4.3† | 7.46±2.41† | 7.4±9.0 | 7.7±10.1 | 0.0055±0.012 | 0 | 11.3±0.3 |
| Ours + IO | – | basic | 2 [1–5] | **79.2**±4.3‡ | **6.78**±2.30‡ | 10.2±10.1† | 11.9±12.5 | 0.58±0.26 | 0 | 153.9±2.6 |
| Ours + IO | – | PET + rig. | 1 [1–5] | 77.3±4.5‡ | 7.01±2.33‡ | 7.0±8.8 | 8.0±10.2 | 0.25±0.25 | 0 | 157.2±2.6 |
| Ours + IO | PET + rig. | PET + rig. | 2 [1–5] | 75.6±4.8 | 7.24±2.38 | 6.2±8.7 | 7.9±9.0 | 0.084±0.089 | 0 | 156.6±2.7 |

the CT overlay (Fig. 5c) and the label residual (e) show misalignment across the whole trunk; after registration (d, f), the residual is reduced to thin bands at organ boundaries. The remaining residual lies mostly at organ boundaries and in the lumbar vertebrae (f), and the lesions keep their shape after warping (h). The Jacobian determinant (g) shows fine-grained local expansion and compression throughout the body, which IO adds to the smooth network field because it has no smoothness term ($\lambda_{\text{smooth}} = 0$), but no folding.

### *4.6 Rigidity Penalties*

We compare rigidity penalties by how well they preserve within-bone distances without sacrificing anatomical alignment or deforming tissue near bone boundaries. [TODO: After the validation sweeps and test runs: curve of Dice vs. distance distortion (validation), test table with bone/soft-tissue alignment, distance distortion, NDV in bone and boundary band, stencil leakage, erosion retention. Leakage on all 560 scans: 21.2 % of stencil reads leave the bone; erosion keeps 35.5 %

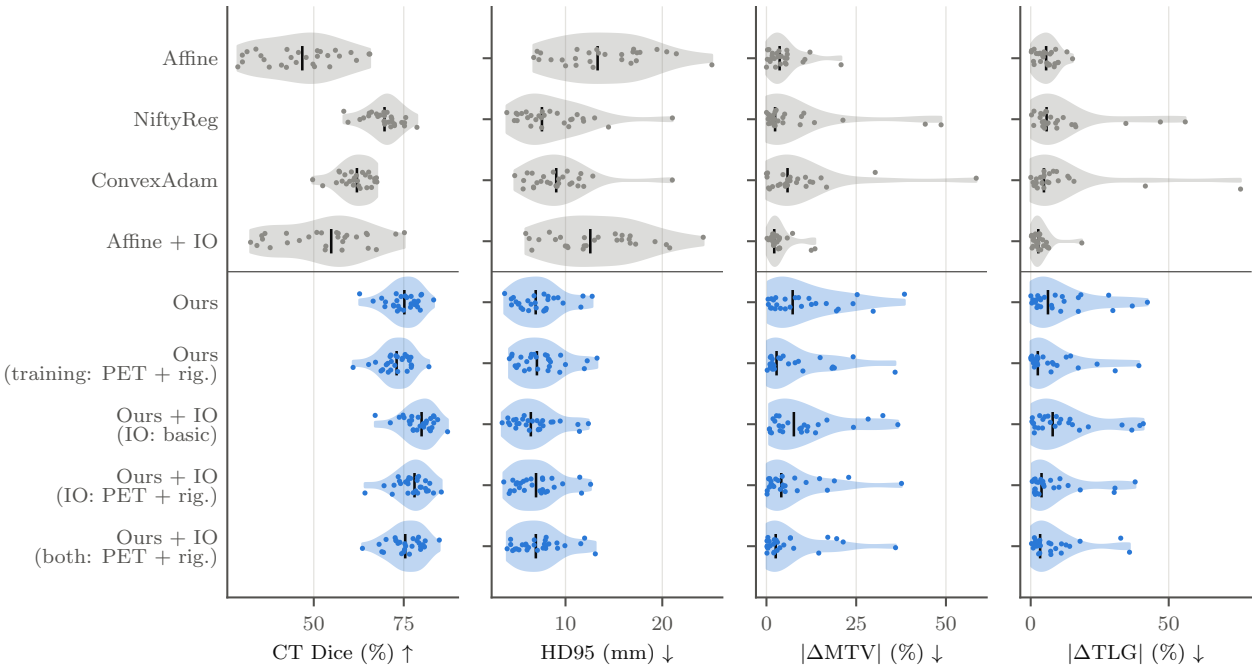



**Fig 4** Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 4).

**Table 5** Rigidity penalties compared (validation set, draft, 26 patients): network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set. Rigidity of bones is measured in three ways: the residual of a rigid fit to each bone (the quantity the rigid-fit penalty minimises), the Staring energy, i.e. the Staring penalty (properness, orthonormality, and affinity) on the whole bone (the quantity the Staring penalty minimises), and the distortion of short (5–15 mm) and long (≥15 mm) within-bone distances (minimised by neither). Mean±SD over patients. $^{\dagger}$/$^{\ddagger}$: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per column, $p < 0.05$).

| | Alignment | | | Rigidity of bones | | | | | Preservation | |
|---|---|---|---|---|---|---|---|---|---|---|
| Rigidity penalty | CT Dice (%) | Bone Dice (%) | Soft-tissue Dice (%) | Rigid-fit residual (mm) | Staring energy | Distortion short (%) | Distortion long (%) | NDV in bone (ppm) | \|ΔMTV\| (%) | \|ΔTLG\| (%) |
| No rigidity | 76.7±4.0 | 77.3±5.3 | 78.3±3.5 | 2.28±0.35 | – | 10.65±1.76 | 4.53±0.74 | 0.6±0.7 | 5.5±6.8 | 6.9±7.9 |
| Staring (×1) | 75.8±3.9$^{\ddagger}$ | 75.8±5.3$^{\ddagger}$ | 78.3±3.6$^{\ddagger}$ | 2.00±0.31$^{\dagger}$ | – | 7.29±1.24$^{\dagger}$ | 3.75±0.58$^{\dagger}$ | 0.0±0.0$^{\dagger}$ | 4.6±5.3 | 6.4±7.6 |
| Staring + erosion (×100) | 76.8±3.9 | 76.9±5.3$^{\ddagger}$ | 79.1±3.6$^{\dagger}$ | 2.11±0.31$^{\dagger}$ | – | 5.51±1.76$^{\dagger}$ | 3.39±0.69$^{\dagger}$ | 3.9±5.4$^{\ddagger}$ | 3.8±4.5 | 5.5±6.3 |
| Per-bone rigid fit (×100) | 75.8±3.9$^{\ddagger}$ | 74.9±5.3$^{\ddagger}$ | 79.1±3.7$^{\dagger}$ | 1.49±0.32$^{\dagger}$ | – | 5.22±1.79$^{\dagger}$ | 2.40±0.85$^{\dagger}$ | 2.1±8.1 | 4.1±4.6 | 5.8±6.5 |

of bone voxels and removes 23 % of bones (ribs: 2.4 % kept, 58 % removed).]

[TODO: Figure 6: wait for the final validation sweep (whole-bone distance distortion), then re-run `figure_rigidity.py` (reads `rigidity_sweep/curve.csv`); currently the first sweep with the interior-only distortion.]

[TODO: Table 5: wait for the final results and switch to the test runs R1–R4 (`rigidity.SOURCE = "test"`), then rerun `table_rigidity.py`; currently a draft on the validation set with the

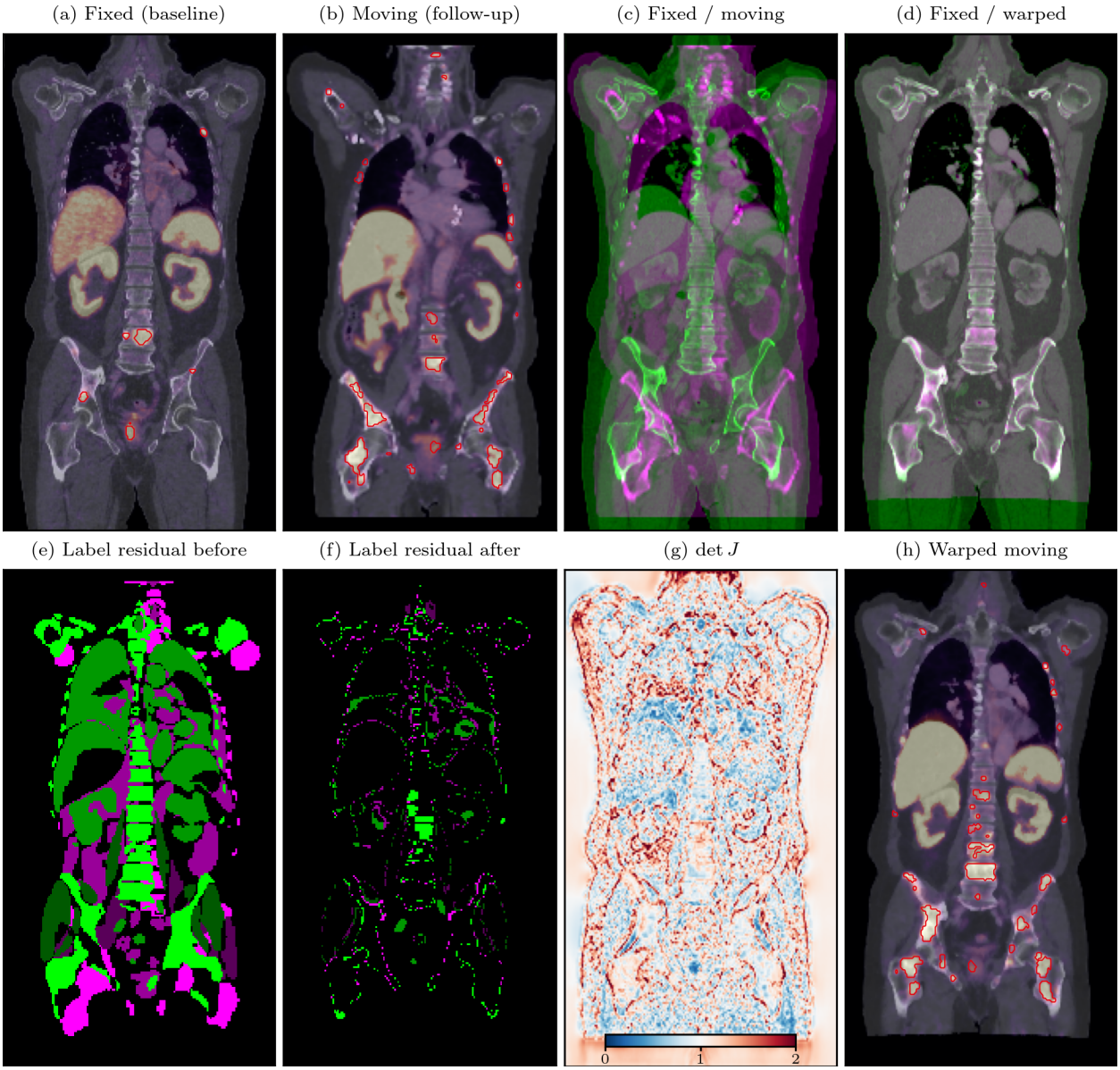



**Fig 5** Test pair with a large change in tumour volume (patient 0146; lesion volume 2400 voxels at baseline, 24191 at follow-up), registered by our full method (PET and rigidity losses in training and IO); one coronal slice. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and ground-truth lesion contours (red). (c, d) CT of the fixed scan (green) overlaid with the moving scan (magenta) before (c) and after (d) registration; grey where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed scan has a label that the moving scan lacks there, magenta where only the moving scan is labelled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the transformation. (h) Warped moving CT with PET and warped lesion contours.

old interior-only distortion and without the Staring energy.]

[TODO: Figure 7: once the final validation sweep is synced, set `SWEEP_DIR` to `rigidity_sweep` and `REFERENCE_CSV` to `evaluation/val/I2.csv` in `figure_rigidity_preservation.py`

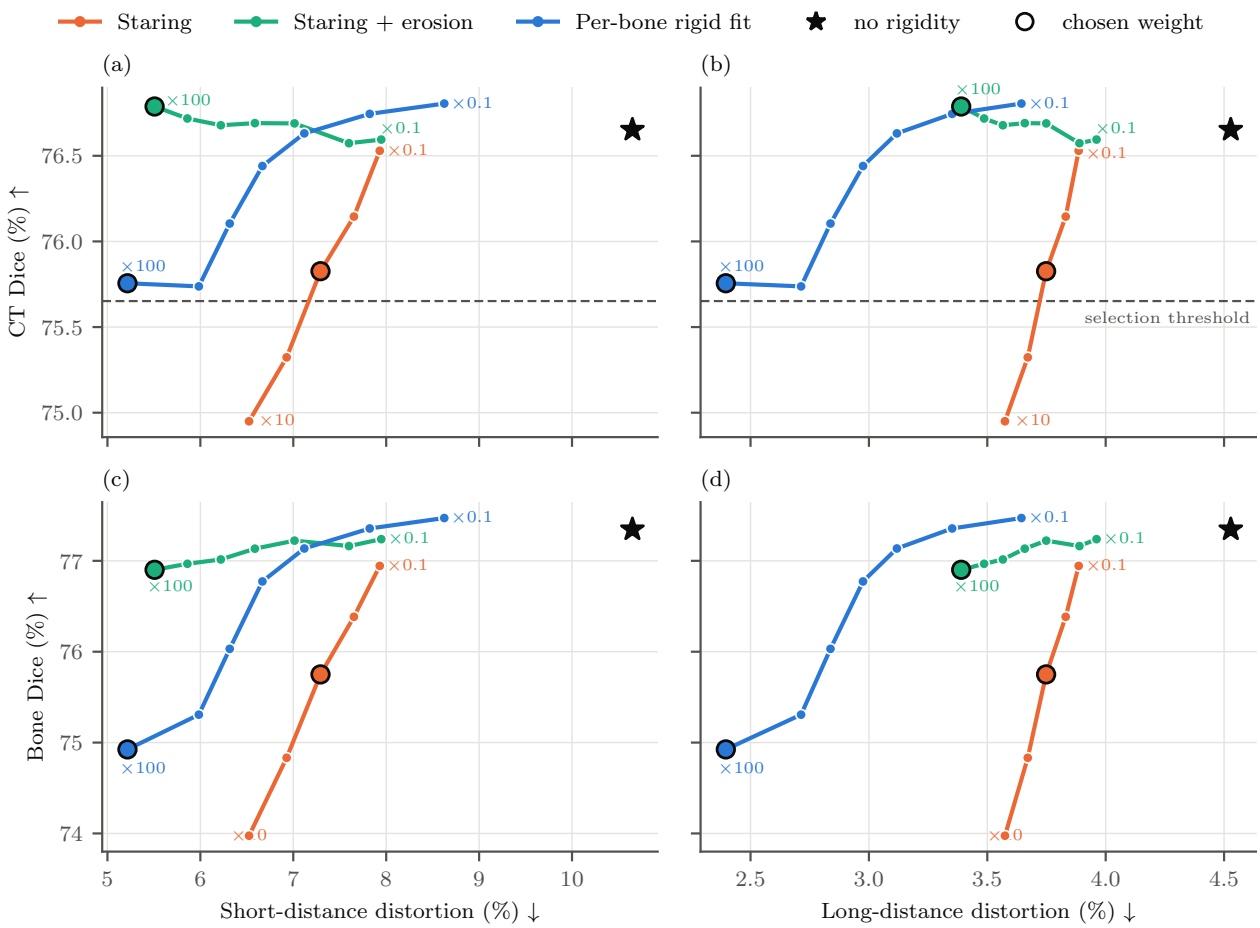



**Fig 6** Rigidity weight sweep on the validation set: network trained with the PET losses, IO with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight (factors at the ends of each line). Top: CT Dice, bottom: bone Dice; left: distortion of short (5–15 mm), right: of long (≥15 mm) within-bone distances, i.e. the mean relative change of the distance between point pairs in the same bone. Star: variant without rigidity; dashed (a, b): selection threshold, one Dice point below it; outlined: weight chosen for the test set (the largest within the threshold, at most ×100). Points: patient means.

and rerun; currently the first sweep.]

### *4.7 Accuracy–Preservation Trade-off*

We vary the PET-loss weights in IO to show how alignment and lesion preservation change together and whether stronger weights resolve the residual error. Figure 8 varies the PET weights in IO on the network trained without the losses. Already the smallest factor ($\times 10^{-4}$) lowered the volume change of the predicted lesion masks from 8.6 % (IO without the PET losses) to 2.2 % and that

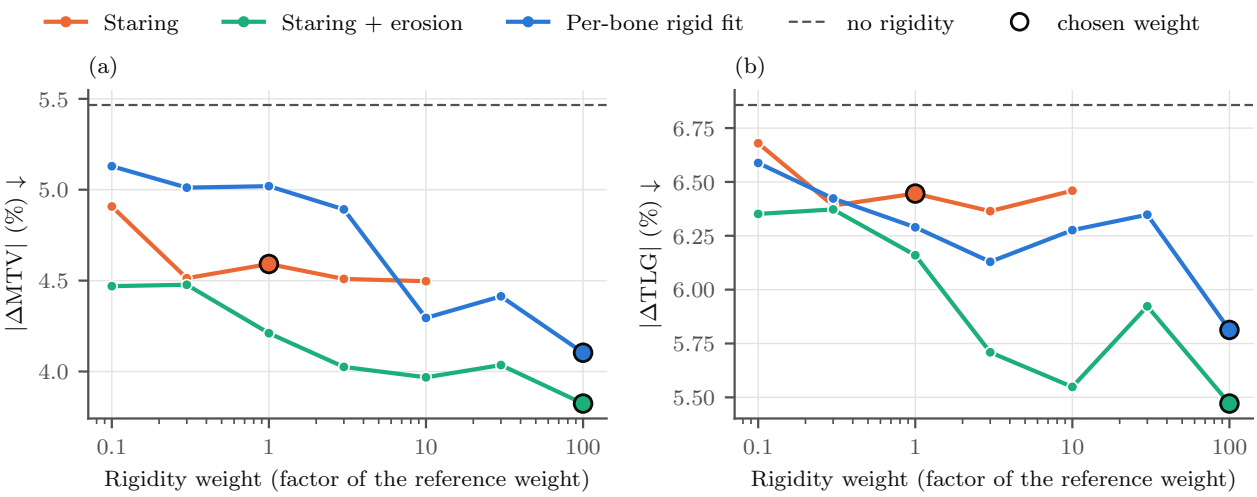



**Fig 7** Effect of the rigidity weight on lesion preservation (validation set): network trained with the PET losses, IO with the PET terms and one rigidity penalty at 0.1 to 100 times its calibrated reference weight. (a) Lesion volume change, (b) uptake change. Dashed: IO without rigidity; outlined: weight chosen for the test set. Points: patient means.

of the ground-truth lesions from 10.2 % to 8.0 %, at a cost of 0.1 Dice points; $|\Delta\mathrm{TLG}|$ fell from 11.8 % to 8.9 %. Larger factors reduced the change of the predicted masks further, to 1.0 % from $\times 0.1$ on, but hardly that of the ground-truth lesions (7.6 % at $\times 0.1$, 7.5 % at $\times 1$), while Dice decreased steadily, to 77.5 % at $\times 0.1$. The factor chosen on the validation set ($\times 0.1$) thus cost 1.6 Dice points more than $\times 10^{-4}$ for 0.4 points less volume change. On the predicted masks, $|\Delta\mathrm{TLG}|$ stayed at about 3 % for all factors, close to the resampling floor of 2.9 %. The same gap between the labels IO optimises and the ground truth appeared for the organs: without the PET losses, IO reached 90.5 % Dice on the predicted organ labels but 79.2 % on the ground-truth labels, starting from 79.0 % and 74.8 % for the network output.

With the ground-truth instead of the predicted lesion masks (black diamonds), IO lowered the change of the ground-truth lesions to 2.1 % for MTV and 3.2 % for TLG at the tuned factor, compared with 7.6 % and 8.7 % with the predicted masks, and was more accurate (78.1 % vs. 77.5 % Dice). At $\times 1$, the values were the same (2.1 %, 3.2 %, 77.7 %): even with correct masks, a larger weight did not improve preservation.

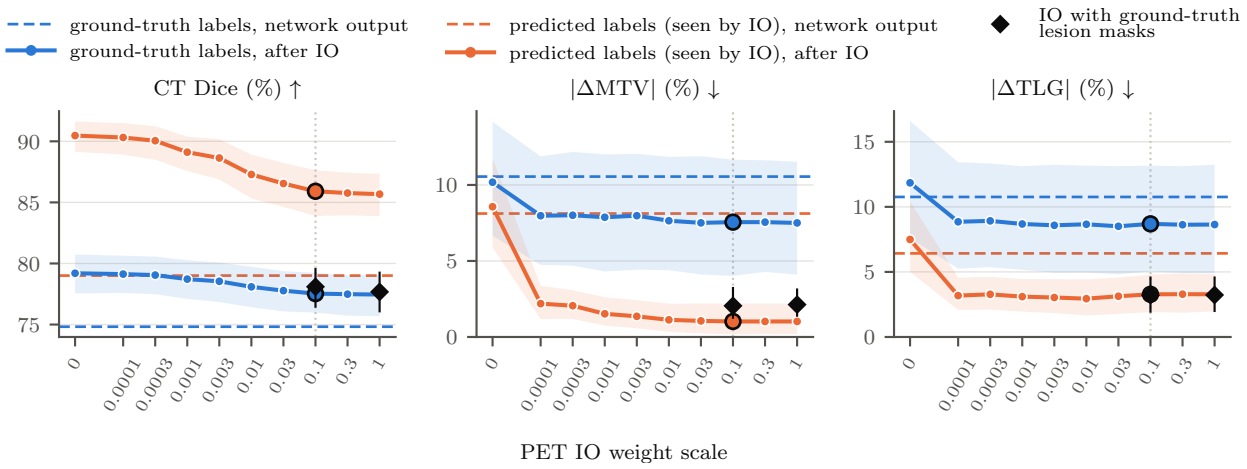



**Fig 8** Effect of the PET loss weight in IO (test set): network trained without PET losses (T1) followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: the PET weights of network training). Each metric on the predicted labels that IO optimises (orange; organs: TotalSegmentator, lesions: nnU-Net) and on the ground-truth labels used for evaluation (blue). Dice over all 28 patients, lesion volume and uptake change over the 26 patients with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Black diamonds: IO with the ground-truth instead of the predicted lesion masks (at $\times 0.1$ and $\times 1$), evaluated on the ground-truth labels. Points: patient means; bands: 95 % patient-bootstrap confidence intervals.

### *4.8 Missed and Detected Lesions*

To locate the remaining registration-induced change, we group ground-truth lesions by their coverage in the predicted mask used by IO (Fig. 9): 214 were missed (6 % of lesion volume), 155 partly covered (6 %), and 1066 covered (88 %).

PET IO reduced the change only of covered lesions, from 7.8 % to 4.9 % for $|\Delta\mathrm{MTV}|$ and from 7.9 % to 5.1 % for $|\Delta\mathrm{TLG}|$ (IO without and with the PET losses at the tuned weights). Missed and partly covered lesions changed as much with the PET losses as without them (missed: 14.9 % and 15.7 %; partly covered: 11.3 % and 11.5 %), since IO does not see them. After PET IO, missed lesions therefore contributed most of the remaining change: 4.8 of 7.6 points of $|\Delta\mathrm{MTV}|$ and 5.7 of 8.7 points of $|\Delta\mathrm{TLG}|$, although they make up only 6 % of the lesion volume. Covered lesions contributed 2.6 and 3.2 points. Scaling the PET weights from $\times 0.1$ to $\times 1$ changed none of these values by more than 0.2 points.

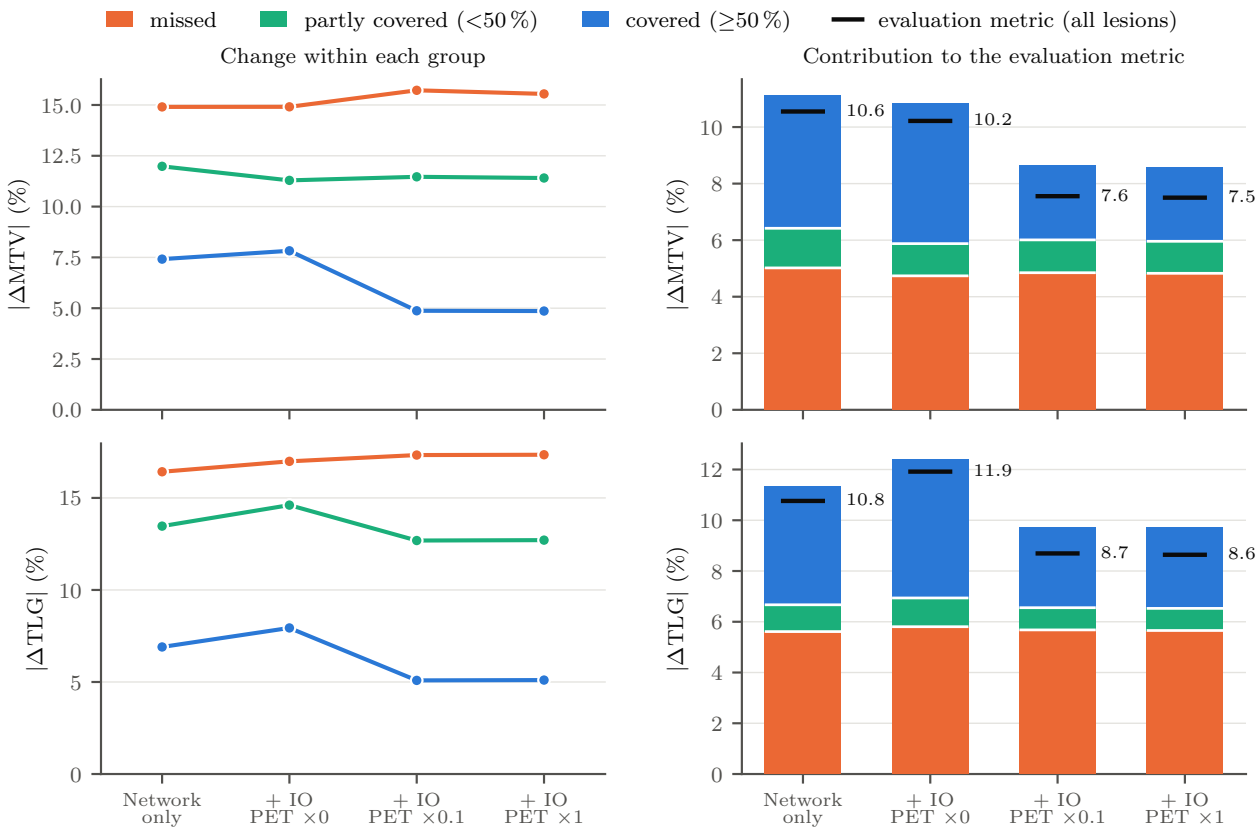



**Fig 9** Lesion volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers: missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 % of volume), covered (1066 lesions, 88 % of volume). Test set; network trained without PET losses, IO with the PET weights scaled by the given factor. Left: change within each group (per scan, the change of the summed volume or uptake of the group's lesions; mean over the patients with lesions in the group). Right: contribution of each group to the evaluation metric (the change per scan if only the group's lesions changed; mean over patients); black: the evaluation metric, below the stacked total because changes of different lesions in a scan partly cancel.

Without the PET losses, IO increased the change of covered lesions over the network output, from 7.4 % to 7.8 % ($|\Delta\mathrm{MTV}|$) and from 6.9 % to 7.9 % ($|\Delta\mathrm{TLG}|$), and the overall $|\Delta\mathrm{TLG}|$ from 10.8 % to 11.9 %; the overall $|\Delta\mathrm{MTV}|$ fell slightly, from 10.6 % to 10.2 %, because missed and partly covered lesions changed less. This agrees with the per-lesion analysis (Sec. 4.3), in which IO without the losses increased the volume change of lesions. With the ground-truth lesion masks in IO, the contribution of the lesions that nnU-Net missed fell from 4.8 to 1.5 points of $|\Delta\mathrm{MTV}|$, and that of the covered lesions from 2.6 to 1.2 points, so that the inaccuracy of the predicted masks explains most of both parts.

## 5 Discussion

The central trade-off is between anatomical alignment and registration-induced change to lesion measurements. PET-preservation losses reduced volume and uptake change at a cost of about two to three Dice points, while the rigidity loss reduced volume change at a smaller alignment cost. When IO was used, applying the losses only during refinement improved alignment over using them in both training and IO. The preservation differences were not statistically significant, but their confidence intervals do not establish that the placements are equivalent. The residual PET change was concentrated in lesions that the predicted masks used by IO missed, showing that the choice of loss placement is only part of the preservation problem.

### *5.1 Choosing Where to Apply the Losses*

Which placement to choose depends on the time budget and on the availability of predicted organ and lesion masks for IO. If IO is affordable, about 2.5 minutes per pair on our GPU, the network can be trained for accuracy alone and the losses applied during refinement. Compared with training and IO with both loss groups, this gained 1.7 Dice points. The paired intervals still allow worse MTV or TLG preservation with IO-only placement, so the result supports an accuracy gain but leaves the preservation trade-off uncertain. Without IO, the network alone registers a pair in 11 s, and training with the losses reduced registration-induced change to lesion volume and uptake. The training losses are therefore not superseded by IO, but they serve a different setting, e.g. fast registration of many follow-up examinations.

One possible explanation for the IO-only result is how each stage locates lesions. During network training, lesion labels define the preservation losses, but the trained registration network receives only images at inference and must respond to lesion-related image cues without a mask.

IO instead uses an explicit mask from a separately trained lesion-segmentation model, which can direct preservation to the predicted lesions in each pair. This could explain why adding the preservation losses during training provides little further measured preservation once IO is applied. The comparison does not test what lesion information the registration network learns, and the segmentation model cannot by itself explain the Dice advantage of IO-only placement: both IO variants use the same predicted masks.

### *5.2 Limits of Lesion Preservation*

On the predicted lesion masks that IO optimises, the PET losses removed almost all volume change (from 8.1 % to 1.0 %), and weights ten times higher than the tuned ones changed none of the metrics. The remaining change on the ground-truth lesions is therefore not a matter of the optimisation or the weights but of the masks: lesions that nnU-Net missed cannot be preserved and contributed most of the remaining volume and uptake change, although they make up only 6 % of the lesion volume. Their weight comes from patients with little tumour volume: the metric is relative per scan and averaged over patients, and in the third of patients with the smallest lesion volume, missed lesions made up 45 % of the volume (18 % and 6 % in the other thirds) and caused 70 % of the missed-lesion contribution. Detecting more lesions at the price of precision is no remedy: on the validation set, IO with a recall-oriented mask (SUV threshold minus physiological organs, combined with nnU-Net), which detected more lesions but had a precision of 0.18, preserved worse than without the PET losses, because preserving the false-positive regions came at the expense of the true lesions. Better lesion segmentation, rather than a different loss weighting, is therefore the most promising way to improve preservation in IO. IO with the ground-truth lesion masks supports this: it lowered the volume change of the ground-truth lesions from 7.6 % to 2.1 % and the uptake

change from 8.7 % to 3.2 %, close to the resampling floor, and was slightly more accurate; the contributions of both missed and detected lesions shrank.

Because IO meets its objective on the predicted masks already at small weights, the PET weight mainly trades accuracy. A factor of $10^{-4}$ of the training-derived weights gave most of the preservation for 0.1 Dice points, whereas the factor chosen by the tuning on the validation set ($\times 0.1$) cost 1.6 points more for 0.4 points less volume change; the tuning grid contained no factor between 0 and $\times 0.1$. Even with the ground-truth masks, a ten times larger weight did not improve preservation.

In IO, the two Jacobian terms carried the volume preservation, whereas the volume term $\mathcal{L}_{\mathrm{MTV}}$ cost 1.3 Dice points and contributed no significant preservation, and the uptake term $\mathcal{L}_{\mathrm{TLG}}$ had no significant effect. A smaller set of terms would likely suffice; we kept all four because they were chosen before the test set was evaluated, and all effects of single terms were below one percentage point.

Aggregate MTV and TLG changes hide how individual lesions are deformed. Small lesions changed most in every condition, partly because a change of one or two voxels is already a large relative change. Without the losses, IO increased the distortion of lesions, most of all of small bone-associated lesions, and bone lesions tended to grow. The rigidity loss acted only on bone-associated lesions, as intended, and the PET losses on both bone and soft-tissue lesions; with both groups of losses, bone lesions changed least, and their systematic growth disappeared.

### *5.3 Interpreting the Evaluation*

The significance-based ranking of the challenge was decided mainly by accuracy: differences in alignment between methods are consistent across patients and reach significance, whereas dif-

ferences in MTV and TLG vary strongly between patients and rarely do. On the validation set, the ranking consequently chose our variant without any of the losses. The ranking also rewards methods that hardly deform: IO directly after the affine pre-registration had the lowest MTV and TLG change of all methods and ranked fifth with 52.7 % Dice, above our network without IO with 74.8 %. Preservation metrics should therefore be read together with accuracy, not on their own. A further effect of the scorer is that warping the PET image changes TLG even for the identity transformation (2.9 % on average); this floor applies equally to all methods but limits how small a TLG change can be reported. Per-lesion changes, which do not cancel between lesions, would complement the aggregate metrics.

Both the accuracy and the preservation of IO are bounded by the labels it uses. On the predicted organ labels, IO raised Dice from 79.0 % to 85.9 %, on the ground-truth labels only from 74.8 % to 77.5 %; for lesions, it preserved the predicted masks almost perfectly but not the missed lesions. Within a challenge whose metrics are computed on segmentations, better segmentation is thus a direct route to better registration scores, for organs as well as for lesions. Outside this setting, label overlap and the change of MTV and TLG are proxies: what matters clinically is whether registration supports downstream tasks such as lesion tracking, response assessment, or dose accumulation. Future work should therefore evaluate registration on such tasks and clinical endpoints rather than optimise further for segmentation-based metrics.

The baseline comparison also illustrates the role of deformation regularity. NiftyReg, tuned on the validation set where it rarely folded, exceeded the folding threshold on 21 % of the test pairs, with three pairs above 0.8 %. The ranking selected its least-regularised setting because accuracy separated the settings more than folding; a setting chosen on the validation set did not transfer to new patients. Our networks with and without IO stayed below the threshold on every pair.

### *5.4 Bone Rigidity*

[TODO: After the rigidity experiments: stencil leakage of finite-difference penalties at bone boundaries (21 % of stencil reads leave the bone), erosion removes most of the skeleton at this resolution (35.5 % of bone voxels kept, ribs 2.4 %), per-bone rigid fit avoids both; results of the comparison.]

### *5.5 Limitations*

The study uses a single cohort with 28 test patients, so small differences, in particular in MTV and TLG, could not be resolved, and generalisation to other scanners and tracers is untested. The images were resampled to about $2.7 \times 2.7 \times 3.3$ mm, at which many lesions consist of few voxels and small lesions are dominated by discretisation. The organ labels used for evaluation are themselves automatic (TotalSegmentator), and the lesion labels used by IO are predictions with a Dice of about 0.5, which, as shown above, limits preservation in IO. All patients, including the test patients, were part of the challenge data with which the earlier version of the method was developed; the test patients were not used for any decision in the present study. We studied one network architecture, and we measured the change caused by registration, not its effect on clinical decisions such as response assessment.

## 6 Conclusion

Lesion-preserving losses reduce the change of PET measurements caused by deformable registration of longitudinal whole-body PSMA PET/CT, at a cost in anatomical alignment. When per-pair instance optimisation is feasible and predicted lesion masks can be obtained, applying the losses only during refinement yielded higher alignment accuracy than using them in training as well. The present test set did not resolve whether PET preservation differs between these placements.

**Table 6** IO tuning on the validation set (network trained with the PET and rigidity losses). Stage 1: learning rate / iterations; stage 2: common scale of the four PET weights relative to the training-derived values. Candidates of a stage are ranked against each other with the significance-based ranking of the challenge (final score $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$); chosen setting in bold. Metrics: patient means. The rigidity weight was fixed at 0.25 (Sec. 3.3).

| Candidate | Rank | Score | Acc. | PET | Reg. | Dice (%) | HD95 (mm) | $\lvert\Delta\mathrm{MTV}\rvert$ (%) | $\lvert\Delta\mathrm{TLG}\rvert$ (%) |
|---|---|---|---|---|---|---|---|---|---|
| *Learning rate and iterations* | | | | | | | | | |
| **0.0125 / 40** | 1 | 25.3 | 0.33 | 0.33 | 0.08 | 76.2 | 6.82 | 5.1 | 3.7 |
| 0.025 / 40 | 2 | 25.1 | 0.38 | 0.29 | 0.08 | 77.0 | 6.83 | 4.1 | 4.7 |
| 0.05 / 20 | 3 | 24.6 | 0.50 | 0.21 | 0.08 | 77.3 | 6.75 | 6.1 | 4.6 |
| 0.025 / 20 | 4 | 23.9 | 0.33 | 0.29 | 0.08 | 76.3 | 6.86 | 4.0 | 4.7 |
| 0.05 / 40 | 5 | 23.1 | 0.71 | 0.12 | 0.08 | 78.1 | 6.71 | 10.0 | 6.8 |
| 0.1 / 40 | 6 | 21.7 | 0.92 | 0.08 | 0.08 | 78.7 | 6.55 | 12.2 | 8.6 |
| 0.1 / 20 | 7 | 20.9 | 0.83 | 0.08 | 0.08 | 78.4 | 6.61 | 13.0 | 8.4 |
| 0.1 / 10 | 8 | 17.6 | 0.54 | 0.08 | 0.08 | 77.4 | 6.74 | 10.4 | 6.6 |
| 0.05 / 10 | 9 | 17.1 | 0.25 | 0.17 | 0.08 | 76.1 | 6.91 | 5.5 | 4.8 |
| 0.0125 / 20 | 9 | 17.1 | 0.17 | 0.25 | 0.08 | 75.1 | 7.16 | 3.9 | 4.8 |
| 0.025 / 10 | 9 | 17.1 | 0.17 | 0.25 | 0.08 | 74.9 | 7.19 | 3.5 | 5.1 |
| 0.0125 / 10 | 12 | 12.0 | 0.08 | 0.21 | 0.08 | 74.0 | 7.26 | 3.9 | 5.4 |
| *PET weight scale (at the chosen learning rate and iterations)* | | | | | | | | | |
| $\times$**0.1** | 1 | 34.4 | 0.86 | 0.21 | 0.14 | 76.4 | 6.85 | 3.6 | 4.6 |
| $\times$0 | 2 | 31.1 | 1.00 | 0.14 | 0.14 | 78.0 | 6.67 | 8.0 | 8.9 |
| $\times$0.25 | 3 | 26.1 | 0.43 | 0.21 | 0.14 | 76.3 | 6.87 | 3.6 | 4.5 |
| $\times$0.5 | 4 | 24.2 | 0.36 | 0.21 | 0.14 | 76.2 | 6.87 | 3.6 | 4.5 |
| $\times$1 | 5 | 22.2 | 0.29 | 0.21 | 0.14 | 76.1 | 6.88 | 3.6 | 4.4 |
| $\times$2 | 6 | 19.8 | 0.21 | 0.21 | 0.14 | 76.1 | 6.88 | 3.7 | 4.4 |
| $\times$4 | 7 | 16.8 | 0.14 | 0.21 | 0.14 | 76.1 | 6.88 | 3.8 | 4.5 |

Without instance optimisation, the training losses reduced the change in lesion measurements. The remaining error was largest for lesions missed by the predicted masks used in instance optimisation, indicating that mask coverage limits the benefit of stronger preservation constraints.

## 7 Appendix A: Tuning of Instance Optimisation

Table 6 lists all candidates of the two IO tuning stages on the validation set (Sec. 3.3) and their ranking.

**Table 7** Signed change of lesion volume and uptake caused by registration (test set, 26 patients with lesions): positive values mean that the warped lesions are larger or take up more. Methods as in Table 4 (Training / IO: losses used; PET + rig.: PET-preservation and rigidity losses). Mean±SD and median over patients.

| | | | ΔMTV (%) | | ΔTLG (%) | |
|---|---|---|---|---|---|---|
| Method | Training | IO | Mean±SD | Median | Mean±SD | Median |
| Affine | – | – | $+2.1 \pm 6.2$ | $+1.2$ | $-2.2 \pm 6.6$ | $-3.5$ |
| NiftyReg | – | – | $+2.6 \pm 14.2$ | $-1.2$ | $-0.7 \pm 17.9$ | $-5.2$ |
| ConvexAdam | – | – | $+8.2 \pm 12.8$ | $+5.8$ | $+5.1 \pm 17.8$ | $+1.9$ |
| Affine + IO | – | PET + rig. | $+1.6 \pm 4.5$ | $+1.9$ | $-0.7 \pm 5.0$ | $+0.7$ |
| Ours | – | – | $+4.4 \pm 14.1$ | $+3.4$ | $+2.7 \pm 15.8$ | $+2.3$ |
| Ours | PET + rig. | – | $+4.2 \pm 11.0$ | $+1.6$ | $+1.9 \pm 12.6$ | $-0.4$ |
| Ours + IO | – | basic | $+2.1 \pm 14.3$ | $+2.6$ | $+1.3 \pm 17.4$ | $+1.2$ |
| Ours + IO | – | PET + rig. | $+2.6 \pm 11.0$ | $+1.3$ | $+1.1 \pm 12.9$ | $+0.5$ |
| Ours + IO | PET + rig. | PET + rig. | $+3.5 \pm 10.1$ | $+0.3$ | $+1.9 \pm 11.9$ | $+1.1$ |

**Table 8** Volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Per scan, the change of the summed volume or uptake of the group's lesions relative to the unwarped moving scan, in %; mean over the patients with lesions in the group.

| | Missed<br>214 lesions, 6 % of volume | | Partly (<50 %)<br>155 lesions, 6 % of volume | | Covered (≥50 %)<br>1066 lesions, 88 % of volume | |
|---|---|---|---|---|---|---|
| | $\lvert\Delta\text{MTV}\rvert$ | $\lvert\Delta\text{TLG}\rvert$ | $\lvert\Delta\text{MTV}\rvert$ | $\lvert\Delta\text{TLG}\rvert$ | $\lvert\Delta\text{MTV}\rvert$ | $\lvert\Delta\text{TLG}\rvert$ |
| Network only | 14.9 | 16.4 | 12.0 | 13.5 | 7.4 | 6.9 |
| + IO, PET ×0 | 14.9 | 17.0 | 11.3 | 14.6 | 7.8 | 7.9 |
| + IO, PET ×0.1 (tuned) | 15.7 | 17.3 | 11.5 | 12.7 | 4.9 | 5.1 |
| + IO, PET ×1 | 15.5 | 17.3 | 11.4 | 12.7 | 4.9 | 5.1 |

## 8 Appendix B: Signed Change of Lesion Volume and Uptake

The main results report the absolute change of MTV and TLG. Table 7 gives the signed change, which shows whether a method systematically enlarges or shrinks the lesions.

## 9 Appendix C: Missed and Detected Lesions

Tables 8 and 9 give the numbers behind Fig. 9.

## 10 Appendix D: Rigidity Penalties

[TODO: Full rigidity table (all measures, incl. the three Staring terms) once the test runs R1–R4 with the whole-bone metrics are in.]

**Table 9** Contribution of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers, to the volume and uptake change of the evaluation (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Contribution of a group: the change per scan if only the group's lesions changed and all other lesions kept their volume and uptake, in %; mean over patients. All: the evaluation metric. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more.

| | Missed<br>214 lesions, 6 % of volume | | Partly (<50 %)<br>155 lesions, 6 % of volume | | Covered (≥50 %)<br>1066 lesions, 88 % of volume | | All lesions<br>1435 lesions | |
|---|---|---|---|---|---|---|---|---|
| | \|ΔMTV\| | \|ΔTLG\| | \|ΔMTV\| | \|ΔTLG\| | \|ΔMTV\| | \|ΔTLG\| | \|ΔMTV\| | \|ΔTLG\| |
| Network only | 5.0 | 5.6 | 1.4 | 1.1 | 4.7 | 4.7 | 10.6 | 10.8 |
| + IO, PET ×0 | 4.7 | 5.8 | 1.1 | 1.1 | 5.0 | 5.5 | 10.2 | 11.9 |
| + IO, PET ×0.1 (tuned) | 4.8 | 5.7 | 1.2 | 0.9 | 2.6 | 3.2 | 7.6 | 8.7 |
| + IO, PET ×1 | 4.8 | 5.7 | 1.1 | 0.9 | 2.6 | 3.2 | 7.5 | 8.6 |

## *Disclosures*

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

## *Code, Data, and Materials Availability*

All experiments use the publicly available training data of the Learn2Reg 2026 PSMAReg challenge,[3,4] which is adapted from the PSMA-PET-CT-Lesions dataset in The Cancer Imaging Archive[5] (DOI: 10.7937/r7ep-3x37). The private test data of the challenge were not used. The patient-level split into training, validation, and test sets is provided with the code. [TODO: Code repository link (training, IO, evaluation, split, and trained weights).]

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*References*

1 Wolfgang P. Fendler, Matthias Eiber, Mohsen Beheshti ... Ken Herrmann, "68Ga-PSMA PET/CT: Joint EANM and SNMMI procedure guideline for prostate cancer imaging: version 1.0," *European Journal of Nuclear Medicine and Molecular Imaging* **44**, 1014–1024 (2017).

2 Learn2Reg Consortium, "Learn2Reg 2026 – PSMAReg: Registration of pre-therapy and follow-up whole-body PSMA PET/CT scans." Codabench challenge (2026). https://www.codabench.org/competitions/15724/.

3 Lasse Hansen, Wiebke Heyer, Christoph Großbröhmer ... Mattias P. Heinrich, "Learn2reg 2024: New benchmark datasets driving progress on new challenges," *Machine Learning for Biomedical Imaging* **3**, 775–791 (2025).

4 Alessa Hering, Lasse Hansen, Tony C. W. Mok ... Mattias P. Heinrich, "Learn2reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation in the era of deep learning," *IEEE Transactions on Medical Imaging* **42**(3), 697–712 (2023).

5 Katharina Jeblick, Stefan O. Schönberg, and others, "A whole-body PSMA-PET/CT dataset

with manually annotated tumor lesions (PSMA-PET-CT-Lesions).” The Cancer Imaging Archive (2024). Version 1.

6 Tom Vercauteren, Xavier Pennec, Aymeric Perchant . . . Nicholas Ayache, “Diffeomorphic demons: Efficient non-parametric image registration,” *NeuroImage* **45**(1, Supplement 1), S61–S72 (2009).

7 B.B. Avants, C.L. Epstein, M. Grossman . . . J.C. Gee, “Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain,” *Medical Image Analysis* **12**(1), 26–41 (2008). Special Issue on The Third International Workshop on Biomedical Image Registration - WBIR 2006.

8 D. Rueckert, L.I. Sonoda, C. Hayes . . . D.J. Hawkes, “Nonrigid registration using free-form deformations: application to breast mr images,” *IEEE Transactions on Medical Imaging* **18**(8), 712–721 (1999).

9 Guha Balakrishnan, Amy Zhao, Mert R. Sabuncu . . . Adrian V. Dalca, “An unsupervised learning model for deformable medical image registration,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, (2018).

10 Adrian V. Dalca, Guha Balakrishnan, John Guttag . . . Mert R. Sabuncu, “Unsupervised learning for fast probabilistic diffeomorphic registration,” in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Alejandro F. Frangi, Julia A. Schnabel, Christos Davatzikos . . . Gabor Fichtinger, Eds., 729–738, Springer International Publishing, (Cham) (2018).

11 Junyu Chen, Eric C. Frey, Yufan He . . . Yong Du, “Transmorph: Transformer for unsupervised medical image registration,” *Medical Image Analysis* **82**, 102615 (2022).

12 Tony C. W. Mok and Albert C. S. Chung, "Large deformation diffeomorphic image registration with laplacian pyramid networks," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2020*, Anne L. Martel, Purang Abolmaesumi, Danail Stoyanov ... Leo Joskowicz, Eds., 211–221, Springer International Publishing, (Cham) (2020).

13 Fryderyk Kögl, Anna Reithmeir, Daniel M. Lang ... Julia A. Schnabel, "PET quantification-preserving registration for longitudinal whole-body PSMA PET/CT," (2026). arXiv:2610.21451.

14 Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer ... Martin Segeroth, "Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence* **5**(5), e230024 (2023).

15 Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl ... Klaus H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods* **18**, 203–211 (2021).

16 Maximilian Rokuss, Balint Kovacs, Yannick Kirchhoff ... Fabian Isensee, "From fdg to psma: A hitchhiker's guide to multitracer, multicenter lesion segmentation in pet/ct imaging," *ArXiv* (2024).

17 Hanna Siebert, Lasse Hansen, and Mattias P. Heinrich, "Fast 3d registration with accurate optimisation and little learning for learn2reg 2021," in *Biomedical Image Registration, Domain Generalisation and Out-of-Distribution Analysis*, Marc Aubreville, David Zimmerer, and Mattias Heinrich, Eds., 174–179, Springer International Publishing, (Cham) (2022).

18 Brian B. Avants, Nicholas J. Tustison, Gang Song ... James C. Gee, "A reproducible evalu-

ation of ants similarity metric performance in brain image registration," *NeuroImage* **54**(3), 2033–2044 (2011).

19 Vincent Arsigny, Olivier Commowick, Xavier Pennec . . . Nicholas Ayache, "A log-Euclidean framework for statistics on diffeomorphisms," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, *Lecture Notes in Computer Science* **4190**, 924–931, Springer (2006).

20 Diederik P. Kingma and Jimmy Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, (2015).

21 Wolfgang Kabsch, "A solution for the best rotation to relate two sets of vectors," *Foundations of Crystallography* **32**(5), 922–923 (1976).

22 S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence* **13**(4), 376–380 (1991).

23 Yihao Liu, Junyu Chen, Shuwen Wei . . . Jerry Prince, "On finite difference jacobian computation in deformable image registration," *International Journal of Computer Vision* **132**, 3678–3688 (2024).

24 Marius Staring, Stefan Klein, and Josien P. W. Pluim, "A rigidity penalty term for nonrigid registration," *Medical Physics* **34**(11), 4098–4108 (2007).

25 Frank Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin* **1**(6), 80–83 (1945).

26 Sture Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics* **6**(2), 65–70 (1979).

27 Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, New York (1993).

28 Marc Modat, Gerard R. Ridgway, Zeike A. Taylor . . . Sébastien Ourselin, "Fast free-form deformation using graphics processing units," *Computer Methods and Programs in Biomedicine* **98**(3), 278–284 (2010). HP-MICCAI 2008.

29 Marc Modat, David M. Cash, Pankaj Daga . . . Sébastien Ourselin, "Global image registration using a symmetric block-matching approach," *Journal of Medical Imaging* **1**(2), 024003 (2014).

# List of Figures

1 Registration pipeline. The follow-up (moving) and baseline (fixed) PET/CT scans are aligned by an affine pre-registration, then by the LapIRN network, which predicts a dense deformation. Instance optimisation (IO) refines this deformation per pair before the moving scan is resampled into the fixed frame. Network training and IO use the same groups of loss terms: alignment, PET preservation, and deformation regularisation (Sec. 2.3). [TODO: Update the figure if the rigidity term or the IO-only variant should appear.]

2 Losses in training and IO versus only in IO (test set, 28 patients). For each loss set, filled: network trained with the losses, then IO with them; hollow: network trained without them (T1), same IO. Grey: neither loss (I1). Points: patient means with 95 % patient-bootstrap confidence intervals.

3 Per-lesion volume change on the test set (1327 lesions of 26 patients), by location (soft tissue / bone-associated, $\geq$50 % of the lesion on bone) and size (small $<$0.5 ml, medium $<$5 ml, large). Median $|\Delta\mathrm{MTV}|$ per lesion; left: network output, right: with instance optimisation (IO) using the same losses. Error bars: 95 % confidence intervals from a patient-cluster bootstrap. Lesions below 5 voxels are excluded.

4 Patient-level results on the test set (28 patients; a patient's pairs are averaged). Violins: kernel density over patients, line: median; each dot is one patient. In brackets: where the PET and rigidity losses (PET + rig.) act; basic: IO with the similarity and regularisation terms only; both: training and IO. Initial alignment is omitted (CT Dice 9.5 %, HD95 44.4 mm; MTV/TLG change 0 by construction, Table 4).

5 Test pair with a large change in tumour volume (patient 0146; lesion volume 2400 voxels at baseline, 24191 at follow-up), registered by our full method (PET and rigidity losses in training and IO); one coronal slice. (a) Fixed and (b) moving CT with PET (SUV 0–8, 35 % opacity) and ground-truth lesion contours (red). (c, d) CT of the fixed scan (green) overlaid with the moving scan (magenta) before (c) and after (d) registration; grey where they agree. (e, f) Organ labels that differ before (e) and after (f) registration: green where the fixed scan has a label that the moving scan lacks there, magenta where only the moving scan is labelled; bright, medium, and dark shades for bones, organs, and muscles. (g) Jacobian determinant of the transformation. (h) Warped moving CT with PET and warped lesion contours.

6 Rigidity weight sweep on the validation set: network trained with the PET losses, IO with the PET terms and one rigidity penalty at $0.1$ to $100$ times its calibrated reference weight (factors at the ends of each line). Top: CT Dice, bottom: bone Dice; left: distortion of short (5–15 mm), right: of long ($\geq$15 mm) within-bone distances, i.e. the mean relative change of the distance between point pairs in the same bone. Star: variant without rigidity; dashed (a, b): selection threshold, one Dice point below it; outlined: weight chosen for the test set (the largest within the threshold, at most $\times 100$). Points: patient means.

7 Effect of the rigidity weight on lesion preservation (validation set): network trained with the PET losses, IO with the PET terms and one rigidity penalty at $0.1$ to $100$ times its calibrated reference weight. (a) Lesion volume change, (b) uptake change. Dashed: IO without rigidity; outlined: weight chosen for the test set. Points: patient means.

8 Effect of the PET loss weight in IO (test set): network trained without PET losses (T1) followed by IO with all PET weights scaled by the given factor (0: PET losses off; 1: the PET weights of network training). Each metric on the predicted labels that IO optimises (orange; organs: TotalSegmentator, lesions: nnU-Net) and on the ground-truth labels used for evaluation (blue). Dice over all 28 patients, lesion volume and uptake change over the 26 patients with lesions in both label sets. Solid: after IO; dashed: network output before IO, the same for every factor. Outlined point and dotted line: operating point tuned on the validation set. Black diamonds: IO with the ground-truth instead of the predicted lesion masks (at $\times 0.1$ and $\times 1$), evaluated on the ground-truth labels. Points: patient means; bands: 95 % patient-bootstrap confidence intervals.

9 Lesion volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers: missed (214 lesions, 6 % of volume), partly covered (155 lesions, 6 % of volume), covered (1066 lesions, 88 % of volume). Test set; network trained without PET losses, IO with the PET weights scaled by the given factor. Left: change within each group (per scan, the change of the summed volume or uptake of the group's lesions; mean over the patients with lesions in the group). Right: contribution of each group to the evaluation metric (the change per scan if only the group's lesions changed; mean over patients); black: the evaluation metric, below the stacked total because changes of different lesions in a scan partly cancel.

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1 Loss weights of Eq. (3) in network training (full-resolution level; the Dice weight is 3, 4, and 5 at the three levels) and in instance optimisation (IO).

2 Effect of the PET and rigidity losses on the test set (28 patients, 38 pairs). Each network is trained with the given losses and evaluated as is (left) and with instance optimisation (IO) using the same losses (right). Mean±SD over patients. $^{\dagger}/^{\ddagger}$: significantly better/worse than "Neither" in the same block (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

3 Contribution of each PET term in IO (test set, 28 patients): network trained without PET losses (T1), IO with all PET terms or with one term removed. Mean±SD over patients. $^{\dagger}/^{\ddagger}$: significantly better/worse than with all terms (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$).

4 Comparison with baselines on the test set (28 patients, 38 follow-up→baseline pairs). Values are mean±SD over patients (a patient's pairs are averaged first). Rank: challenge significance ranking, with the 95 % interval of the rank over 1000 patient bootstrap samples in brackets, over all methods except Initial, which is a reference (its MTV/TLG change is 0 by construction because nothing is resampled). Training / IO: PET and rigidity losses (PET + rig.) used in training and in instance optimisation (IO); IO "–": no IO, "basic": IO with the similarity and regularisation terms only. †/‡: Ours + IO with PET + rig. in training and IO (our full method) is significantly better/worse than this method (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected within each column, $p < 0.05$; NDV tested after the 0.005 % equivalence threshold). $>$ thr.: pairs with %NDV above 0.005 %. Runtime per pair on one NVIDIA RTX A6000, including the affine step. Affine + IO: our instance optimisation directly after the affine pre-registration, without the network. Bold: best mean among ranked methods.

5 Rigidity penalties compared (validation set, draft, 26 patients): network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set. Rigidity of bones is measured in three ways: the residual of a rigid fit to each bone (the quantity the rigid-fit penalty minimises), the Staring energy, i.e. the Staring penalty (properness, orthonormality, and affinity) on the whole bone (the quantity the Staring penalty minimises), and the distortion of short (5–15 mm) and long ($\geq$15 mm) within-bone distances (minimised by neither). Mean$\pm$SD over patients. $^{\dagger}$/$^{\ddagger}$: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per column, $p < 0.05$).

6 IO tuning on the validation set (network trained with the PET and rigidity losses). Stage 1: learning rate / iterations; stage 2: common scale of the four PET weights relative to the training-derived values. Candidates of a stage are ranked against each other with the significance-based ranking of the challenge (final score $100 \cdot \mathrm{acc}^{0.4}\,\mathrm{PET}^{0.4}\,\mathrm{reg}^{0.2}$); chosen setting in bold. Metrics: patient means. The rigidity weight was fixed at 0.25 (Sec. 3.3).

7 Signed change of lesion volume and uptake caused by registration (test set, 26 patients with lesions): positive values mean that the warped lesions are larger or take up more. Methods as in Table 4 (Training / IO: losses used; PET + rig.: PET-preservation and rigidity losses). Mean$\pm$SD and median over patients.

8 Volume and uptake change of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Per scan, the change of the summed volume or uptake of the group's lesions relative to the unwarped moving scan, in %; mean over the patients with lesions in the group.

9 Contribution of ground-truth lesions, grouped by how much of each lesion the predicted lesion mask seen by IO covers, to the volume and uptake change of the evaluation (test set; network trained without PET losses, IO with the PET weights scaled by the given factor). Contribution of a group: the change per scan if only the group's lesions changed and all other lesions kept their volume and uptake, in %; mean over patients. All: the evaluation metric. Since changes of different lesions in a scan partly cancel, the contributions add up to slightly more.